EFY TESTED
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Solar Pooja Lamp That Runs Nonstop

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The oil in lamps that are traditionally lit in a pooja room has to be topped up from time to time. The solar lamp described here can provide light nonstop for many years without any intervention.

In this system, a small solar photovoltaic (PV) panel turns on the LEDs during daytime and charges two supercapacitors during the day. In the evening, when the daytime LEDs turn off, another set of LEDs (joule thief) is turned on using energy stored in the supercapacitors.

As supercapacitors can be charged and discharged any number of times, the whole system can work nonstop for 15 years or more. The author's prototype with joule thief LEDs on is shown in Fig. 1.

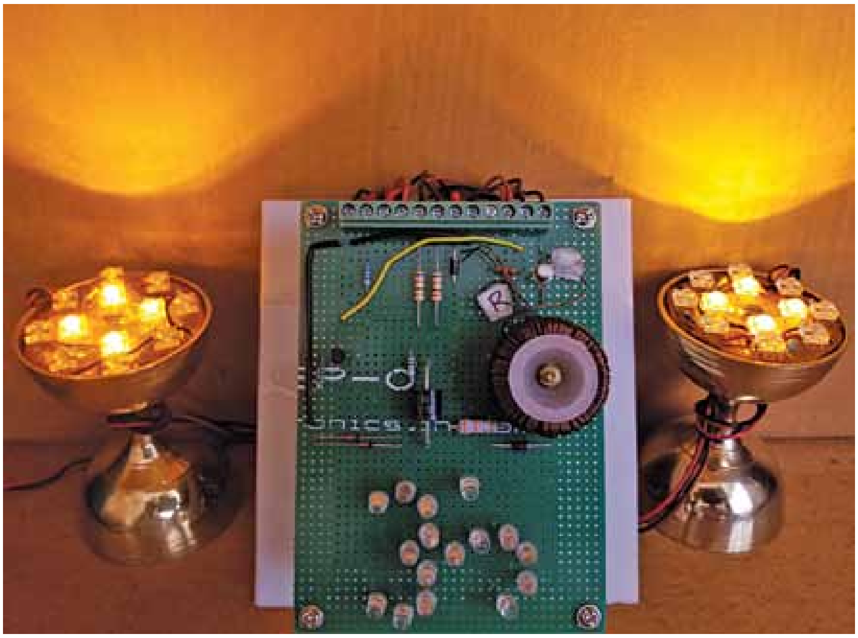

Fig. 1: Author's prototype with joule thief LEDs on

Circuit and working

The circuit diagram of solar pooja lamp using LEDs is shown in Fig. 2. The complete system comprises solar lamp LEDs, supercapacitor charging circuit, and joule thief circuit.

The solar daylamp uses a 10W-peak solar PV panel whose specifications are:

Voltage at maximum power (Vmp) = 17.5V

Current at maximum power (Imp) = 0.58A

The daylamp has two arrays of eight connected yellow high-brightness LEDs each. These LEDs (LED1 through LED16) are assembled on two octagonal PCBs as shown in Fig. 3. LED1 through LED8 are soldered on left side lamp (LL) PCB and LED9 through LED16 are soldered on right side lamp (LR) PCB. The respective current-limiting resistors R1 through R8 are also mounted on these PCBs. The PCBs can be mounted on a brass lamp that is easily available in the market.

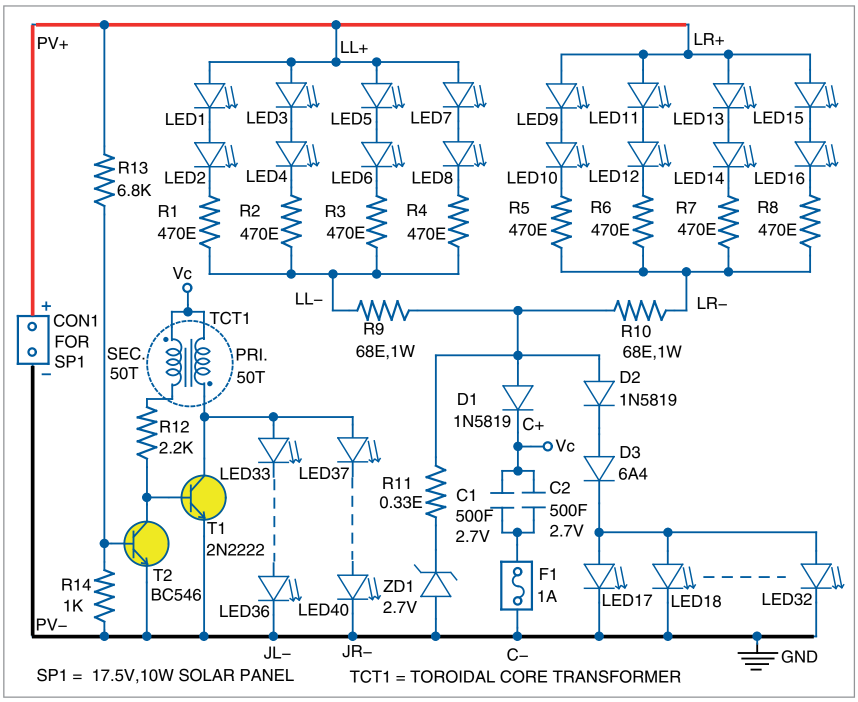

Fig. 2: Circuit diagram of pooja lamp

The positive connections from